MAPS OUT OF A QUASI-COHERENT OBJECT ARE COMPUTED BY THE DERIVED HOM OF GLOBAL SECTIONS OVER AN OBJECT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is written out in full: the reduction to a final object, the presentation of the quasi-coherent object, and the concluding adjunction.

Editorial note. The label lemma-QC-hom-out-of names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

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Allowed standard prerequisites (editorial).

- Localisation to an object of the site. Same chapter, the reduction of statements about the site to the localisation at an object. Supplies the passage from a general U to the case of a final object.
- Description of quasi-coherent objects over a final object. Same chapter, the lemma identifying a quasi-coherent object with the pullback of its global sections when the site has a final object. Supplies the presentation K = Lf^*P with P = R Gamma(U, K).
- Adjunction between the pullback and the pushforward along the morphism to the point. Same chapter, the adjunction Lf^* left adjoint to Rf_* = R Gamma(U, -). Supplies the displayed identification of the two Hom groups.

Source reference. Original reference 0H0R: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 11161–11161. Prerequisite pointer invoked at proof line(s) 11186; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *In the situation above, suppose that K is an object of $QC(\mathcal{O})$ and M arbitrary in $D(\mathcal{O})$. For every object U of $\mathcal{C}$ we have*

$$\mathrm{Hom}_{D(\mathcal{O}_U)}(K|_U, M|_U) = R\,\mathrm{Hom}_{\mathcal{O}(U)}(R\Gamma(U, K), R\Gamma(U, M))$$

Proof. We may replace $\mathcal{C}$ by $\mathcal{C}/U$. Thus we may assume $U = X$ is a final object of $\mathcal{C}$. By Lemma 0H0R we see that $K = Lf^*P$ where $P = R\Gamma(U, K) = R\Gamma(X, K) = Rf_*K$. Thus the result because Lf^* is the left adjoint to $Rf_*(-) = R\Gamma(U, -)$. □

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